MIL-E-1/1332B (USAF)
27 January 1971
SUPERSEDING
MIL-E-1/1332A (USAF)
12 May 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRIODE, POWER

TYPE 6950 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Shielded-grid beam triode, pulse amplifier transmitting

PHYSICAL CHARACTERISTICS:

Mounting position:	Tube axis vertical, filament terminal down.
Cathode:	Multistrand thoriated-tungsten filament.
Connections and dimensional outline:	See figures 1 and 2.
Weight:	204 pounds nominal.

ABSOLUTE RATINGS:
(Anode pulsed Class B amplifier)

Parameter:	Ef	eb	Ec1	ib	ic1	Ib	Ic1	Pi	Pp
Unit:	V	kv	Vdc	a	a	Adc	Adc	kW	kW
Note:	1	11	-----	--	---	---	---	---	
Maximum:	7.8	40	-1000	92	5	5.7	0.3	230	150
Minimum:	7.0	--	-----	--	---	---	---	---	---
TEST CONDITIONS:	7.3	--	-----	--	---	---	---	---	---

ABSOLUTE RATINGS:
(Anode pulsed Class B amplifier)

Parameter:	tk	tp	F	Cooling	Fault Protection
Unit:	sec	μs	MHz	--	--
Note:	1	10	2	3	4
Maximum:	-	2500	195	--	--
Minimum:	30	----	---	--	--
TEST CONDITIONS:	--	----	---	See note 3	--

GENERAL:

Qualification:	See note 5
Holding period:	(t = 96 hours)

1/ Formerly 6950/RCA-2039

(B) Complete Revision

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PAR. NO.	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1155	Anode water pressure differential	No voltages See note 6	0.65	II	PPD	---	33	psi
1155	Beam former pressure differential	No voltages See note 6	0.65	II	BFPD	---	12	psi
1261	Electrode voltage (grid)	Eb = 20k Vdc Ec1/Ib = 0.25 Adc	0.65	II	Ec1	---	-450	Vdc
1301	Electrode current (filament)		0.65	II	If	1060	1220	A
1256	Electrode current (anode)	Eb = -22.5 Vdc Ec1/Ic1 = 2.0 Adc t = 5 minutes See notes 7 and 12	0.65	II	Ib	---	-20	µAdc
1316	Amplification factor	Eb/Ib = 10 Adc Ec1 (1) = 0 Vdc Ec1 (2) = -100 Vdc µ = Eb (2) - Eb (1) /100	0.65	II	µ	46	68	---
----	Power output (1)	Ef = 7.3 to 7.8 V eb = 30 kv tp = 2000 µs t = 8 hour; Peak driver power = 120 kw (max) See note 8	0.65	II	Po(1)	1.5	---	Mw
----	Power output (2)	Eb = 20k Vdc Ec1/Ib = 50 A Ec = 2 Ec1 F = 425 ± 25 kHz t = 30 minutes See note 8	0.65	II	Po (2)	700	---	kW
	Quality conformance inspection, part 2							
----	Input resonant frequency	See note 9	---	---	Fi	122	152	MHz
----	Output resonant frequency	See note 9	---	---	Fo	138	168	MHz
1105	Permanence of marking	-------	---	---	---	---	---	---
	Quality conformance inspection, part 2							
----	Service life guarantee	Within absolute ratings	---	---	t	1000	---	hours
----	Service life end points	Power output (1) Power output (2)	--- ---	--- ---	Po(1) Po(2)	1.2 600	--- ---	Mw kW

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NOTES:

1. Filament Voltage

Time (tk) begins after filament voltage is raised to operating value. To avoid undue thermal stresses in the filament, it is recommended that the filament voltage be raised to operating value in not less than 30 seconds. Filament surge current shall never exceed 1700 amperes, even momentarily.

2. Frequency

Operation above 195 MHz is possible at derated conditions.

3. Cooling
Ratings:

Water cooling of the beam-forming cylinder, each grid-terminal connector and the anode is required. The water flow shall start before application of any voltages and preferably should continue for several minutes after removal of all voltages. Interlocking of the water flow for each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate water flow. The use of distilled or de-ionized water is essential. Water flow requirements are as follows:

	Min. gpm	Typical gpm	Max. Pressure Differential (psi) for Typical Flow (See Note Below)	Max. Pressure (psig)
Anode inlet (for Pp = 150 kw)	80	88	40	100
Beam-Forming Cylinder	7	8	16	50

Note: Measured directly across the coolant element.

Maximum or minimum temperatures shall be as follows:

a. Water (at any outlet): 70°C max.
b. Ceramic-bushing: 150°C max.
c. Metal-surface: 100°C max.
d. Storage Temperature: -65°C min.

(Water cooled elements shall be free of water before storate or shipment damage from freezing).

Test Conditions:

The water flow to the anode cooling connection shall be 80 gpm minimum. The water flow to the beam-forming cylinder connection shall be 7gpm minimum. Water flow shall start before the application of any voltages and shall continue for at least 10 seconds after removal of all voltages. The outlet water temperature shall not exceed 70°C.

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4. Fault Protection

High speed electronic protective devices shall be used to prevent energy from the anode power supply and the grid-No. 1 power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the various pulse sources supplying the tube within 20 microseconds following the occurrence of a fault in any single pulse. All pulses subsequent to a "fault" pulse will not be applied until the device is reset. Minimum reset time shall be 0.1 second. (A fault is a condition in which the peak pulse currents rise 20 percent or more above the normal peak current level.)

5. Qualification

All reference to qualification test in MIL-E-1 shall be deleted and replaced by preproduction test. All tests specified herein shall be conducted on a preproduction sample which shall be representative of the production tubes. The first article test program is designed to demonstrate the manufacturer's capability to meet the performance specification on the tube. Therefore, the first article requirement is applicable only on the first procurement from any supplier. On all subsequent procurements, the first article requirement will not apply.

6. Water Pressure Differential

The water flow to the anode cooling connection shall be 80 gpm. The water flow to the beam-former cooling connection shall be 7 gpm.

7. Anode Current

After the specified time, record anode current as -1 Ib. Then remove grid-No. 1 voltage and record anode current as -2 Ib. Anode current -1Ib=(-1Ib) minus (-2Ib).

8. Power Output
Only one Power Output Test will be performed. Power Output (1) shall be performed unless authority is granted by the procuring activity to perform Power Output (2) therefor.

Power Output (1)

The continuous pulsed power output for the specified period shall be defined as the total accumulated operating time at Power Output (1) test conditions, less the outage time due to equipment malfunction or trip-out. During the operational time, the tube shall not show evidence of "slumping" or "cave-in". This test shall be conducted at a frequency (between 175 MHz and 195 MHz) specified by the procuring activity.

Power Output (2)

The tube shall be tested in a 425 kHz amplifier test set. The grid drive, coupling and output circuit loading shall be adjusted to obtain specified operating conditions. ecl max. = 1700V. Power output shall be measured across a water-cooled resistive load. Power output shall be determined by recording the inlet and outlet water temperature of the load and calculated as follows:

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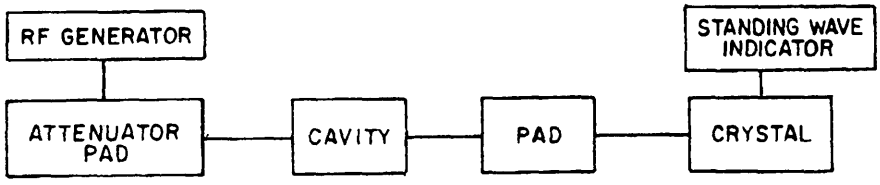
NOTES: (Cont'd)

RFPo (kilowatts) = 0.264n (to - ti)

Where "n" is the water flow in gpm; "to" is the outlet water temperature and "ti" is the inlet water temperature in °C.

9. Input and Output Resonant Frequency

The RF frequency generator shall be adjusted to obtain a resonant peak on the standing wave indicator, and frequency shall be measured as in the block diagram shown below: (Be sure tube is making good contact with cavity.)



Input:

Three input resonant cavities, each mounted at the designated location of the tube, shall be in accordance with figures 3, 4 and 5.

Output:

Two identical output resonant cavities, each mounted at opposite ends of the tube, shall be in accordance with figure 6.

10. Pulse Duration

On time (tp) is defined as the sum of the durations of all individual pulses which occur during any 40,000 microsecond interval.

11. Anode Voltage

Pressurization of the output and input cavities may be required to prevent corona or external flashover at the ceramic bushings.

12. Holding Period

Test to be performed at the conclusion of the holding period.

Custodian:
Air Force - 80

Review activities:
Air Force - 11, 17
DSA-ES

Preparing activity:
Air Force - 80

(Project 5960-F826-5)

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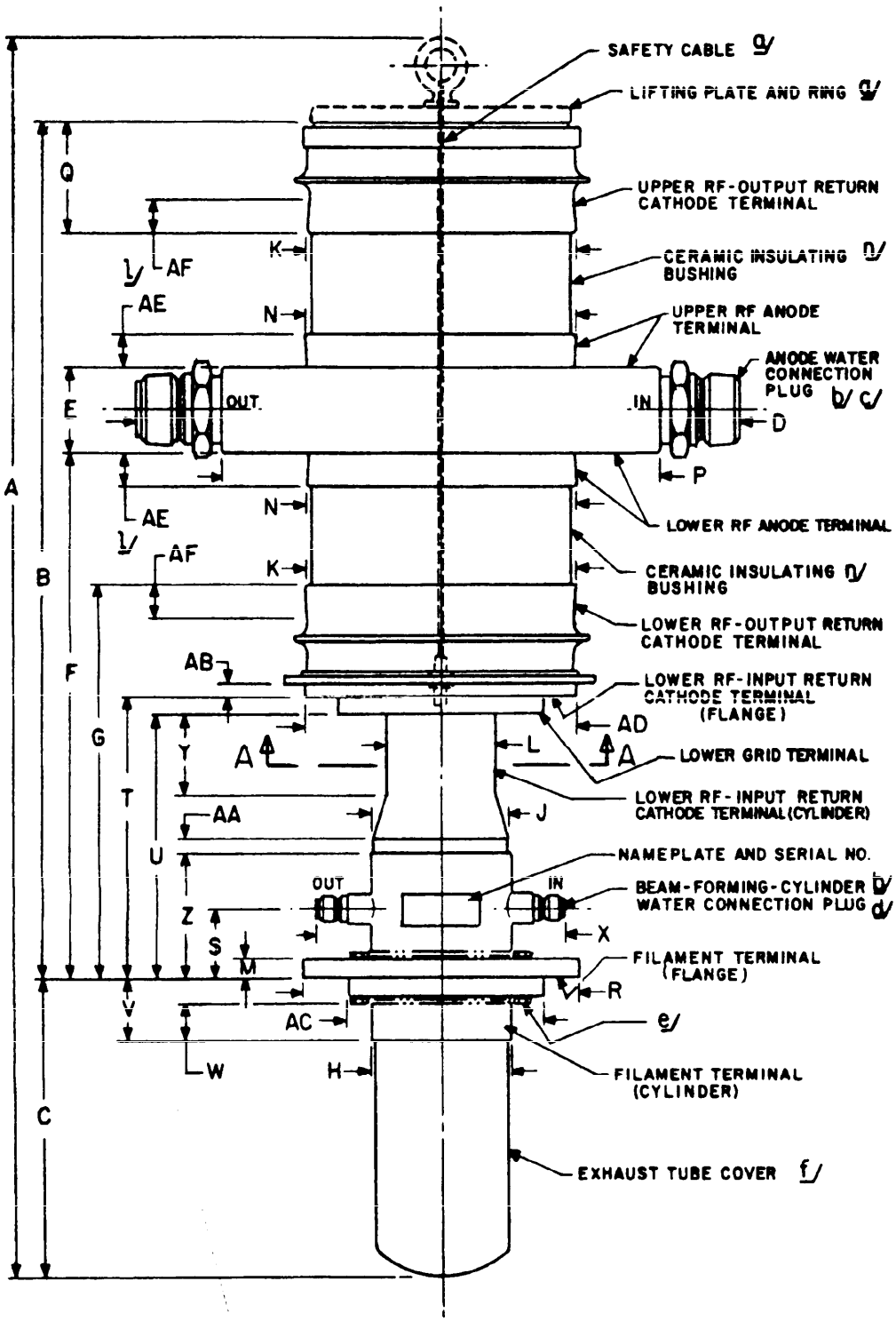


FIGURE 1. Outline drawing Electron Tube type 6950

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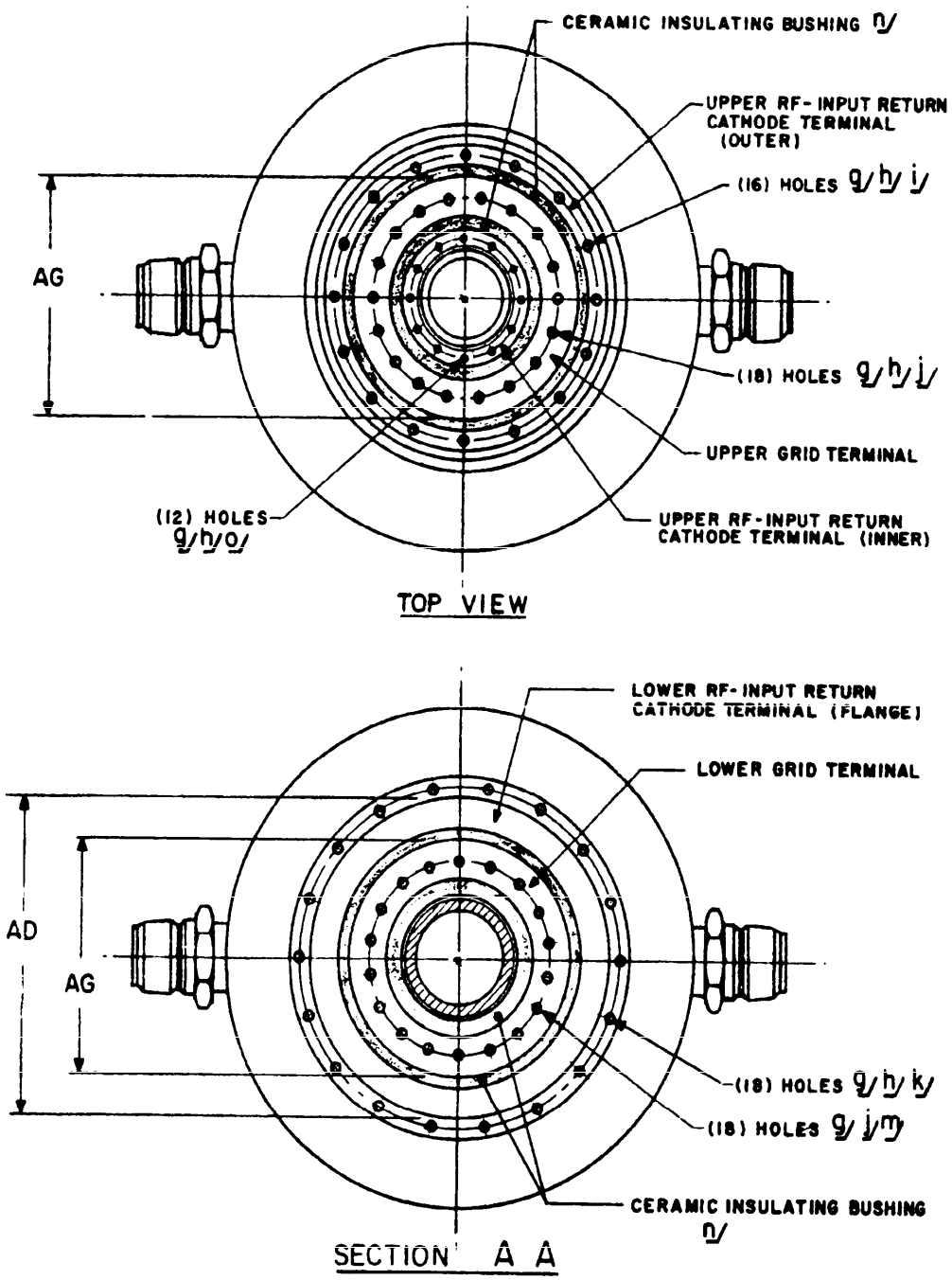


FIGURE 2. Outline drawing Electron Tube type 6950

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DIMENSIONS (FIGURES 1 & 2)									
LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART I					Q	3.16	3.32	80.3	84.3
B	25.10	25.58	637.6	649.7	R DIA	8.78	8.90	223.0	226.0
C	-	9.24	-	294.7	S	1.94	2.18	49.3	55.3
G	11.48	11.88	291.6	302.7	T	8.26	8.50	209.8	215.9
N DIA	8.52	8.72	216.4	221.4	U	7.76	8.00	197.1	203.2
					V	1.82	2.06	46.2	52.3
PREPRODUCTION					W	.8	-	20.3	-
A	-	37.24	-	945.9	X	-	8.4	-	213.3
D	-	20.50	-	520.7	Y	2.00	-	50.8	-
E	2.46	2.54	62.48	64.51	Z	3.62	3.78	91.95	96.01
F	15.58	15.74	395.7	399.8	AB	.34	.42	8.64	10.66
H DIA	4.44	4.50	112.78	114.30	AC DIA	-	6.12	-	155.4
J DIA	4.28	4.38	108.71	111.25	AD DIA	8.76	8.88	222.5	225.6
K DIA	8.52	8.72	216.4	221.4	AE	.88	-	22.4	-
L DIA	3.45	3.55	87.6	90.1	AF	1.00	-	25.4	-
M	.50	.62	12.7	15.7	AG DIA	6.50	6.62	165.1	168.1
P DIA	-	14.12	-	358.6	AA	.38	-	9.65	-

NOTES:

a. Remove the lifting plate and safety cable before operating tube. Keep them for future tube handling:

b. Inlet water connections (IN) are both on same side of tube and to the right when tube is viewed with nameplate toward observer.

c. This plug is removable and is Hansen Mfg. Co., ( ) No. 12T45, 1-1/2 inch (38.1 mm) or equivalent.

d. This plug is not removable and is Murphy G. W. Industries, Inc., Pneumatic Division (95138) No. E4M4, 1/2 inch (12.7 mm).

e. Do not tamper with these 9 or 18 bolts. The number of bolts is dependent on construction.

f. Do not use for supporting weight of tube. Do not remove. Make no connection.

h. The planes of the upper RF-input return cathode terminals (inner and outer) and the upper grid terminal are parallel with 0.060 inch (1.52 mm) and coincident within 0.090 inch (2.29 mm). Run-out between the bolt circles of the upper terminal will be within 0.12 inch (3.0 mm).

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NOTES:

i. Sixteen (16) holes, tapped .250-28 UNF-1B, 0.31 inch (7.9 mm) minimum depth, equally spaced within 0.025 inch (.64 mm) on a bolt circle diameter of 7.857 ± 0.030 inch (199.6 ± .5 mm).

j. Eighteen (18) holes tapped .250-28 UNF-1B, 0.38 inch (9.7 mm) minimum depth, equally spaced within 0.025 inch (.64 mm) on a bolt circle diameter of 5.450 ± 0.025 inch (138.43 ± .64 mm).

k. Eighteen (18) holes equally spaced within 0.025 inch (.64 mm) on a bolt circle diameter of 9.50 ± 0.03 inch (241.3 ± 5 mm) for .250 inch (6.35 mm) diameter bolts.

l. This area is subject to a maximum taper of 0.06 inch to the inch. The maximum diameter along this taper sill be on the end toward the ceramic.

m. Run-out between bolt circles of the lower terminals will be within 0.12 inch (3.0 mm).

n. A potting compound (silicone rubber, or equivalent) shall be applied in the space formed at the junction of each of the following metal terminals and its associated ceramic bushing:

(1) The upper rf-output return cathode terminal.

(2) The upper rf-anode terminal.

(3) The lower rf-output return cathode terminal.

(4) The lower rf-anode terminal.

(5) The rf-input return cathode terminal flange.

(6) The upper grid terminal.

(7) The lower rf-input return cathode terminal cylinder.

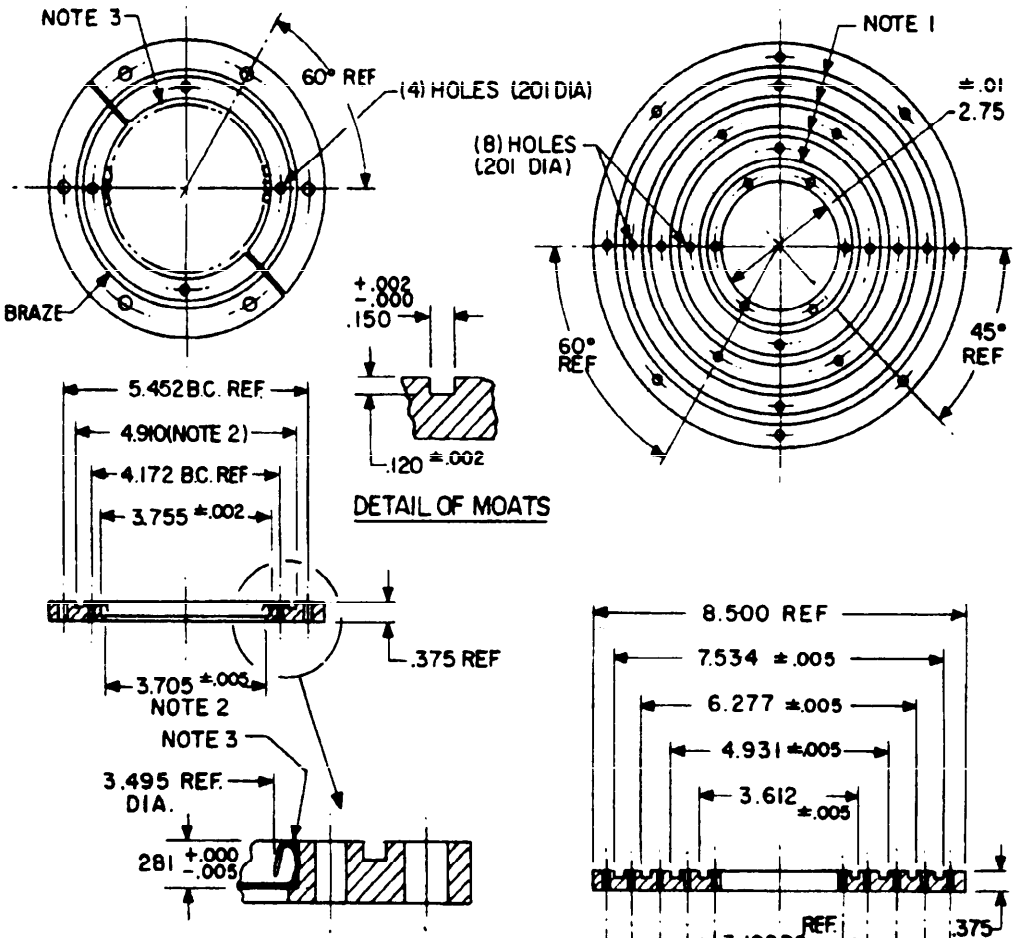
(8) The lower grid terminal.

o. Twelve (12) holes threaded 6-32 UNC, 0.31 inch (7.9 mm) minimum depth, equally spaced within 0.03 inch (0.8 mm) on a bolt circle diameter of 3.10 ± 0.05 inch (78.7 ± 1.3 mm). First thread to start 0.065 inch (16.51 mm) maximum depth.

p. Metric equivalents are given for general information only and are based on 1 inch = 25.4 mm.

q. One tube in each group of ten tubes, starting with the first tube delivered shall be chosen by the cognizant government representative for testing under quality conformance inspection tests. Should any failure occur, the government representative may, at his discretion, require that all further tubes be subjected to the test in which failure occurred until such time that performance, in his opinion, has demonstrated that this testing is no longer necessary.

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NOTES:-

1. ASSEMBLY CONSISTS OF TWO HALVES.
2. RUN-OUT BETWEEN DIAMETERS INDICATED NOT TO EXCEED .002
3. SPRING CONTACT RING, SUCH AS CAT. NO. 97-135, INSTRUMENT SPECIALITIES CO. INC. LITTLE FALLS, N. J. (CODE 30817) MUST INSURE CONTINUOUS CONTACT.
4. MOTE FOR BRAIDED CONTACT. THE BRAIDED RING, SUCH AS CAT NO. 10-118, ELECTRONICS DIVISION, METAL TEXTILE CORP., ROSELLE, N.J. (CODE 82805). MUST INSURE CONTINUOUS CONTACT.
5. DIMENSIONS ARE IN INCHES.

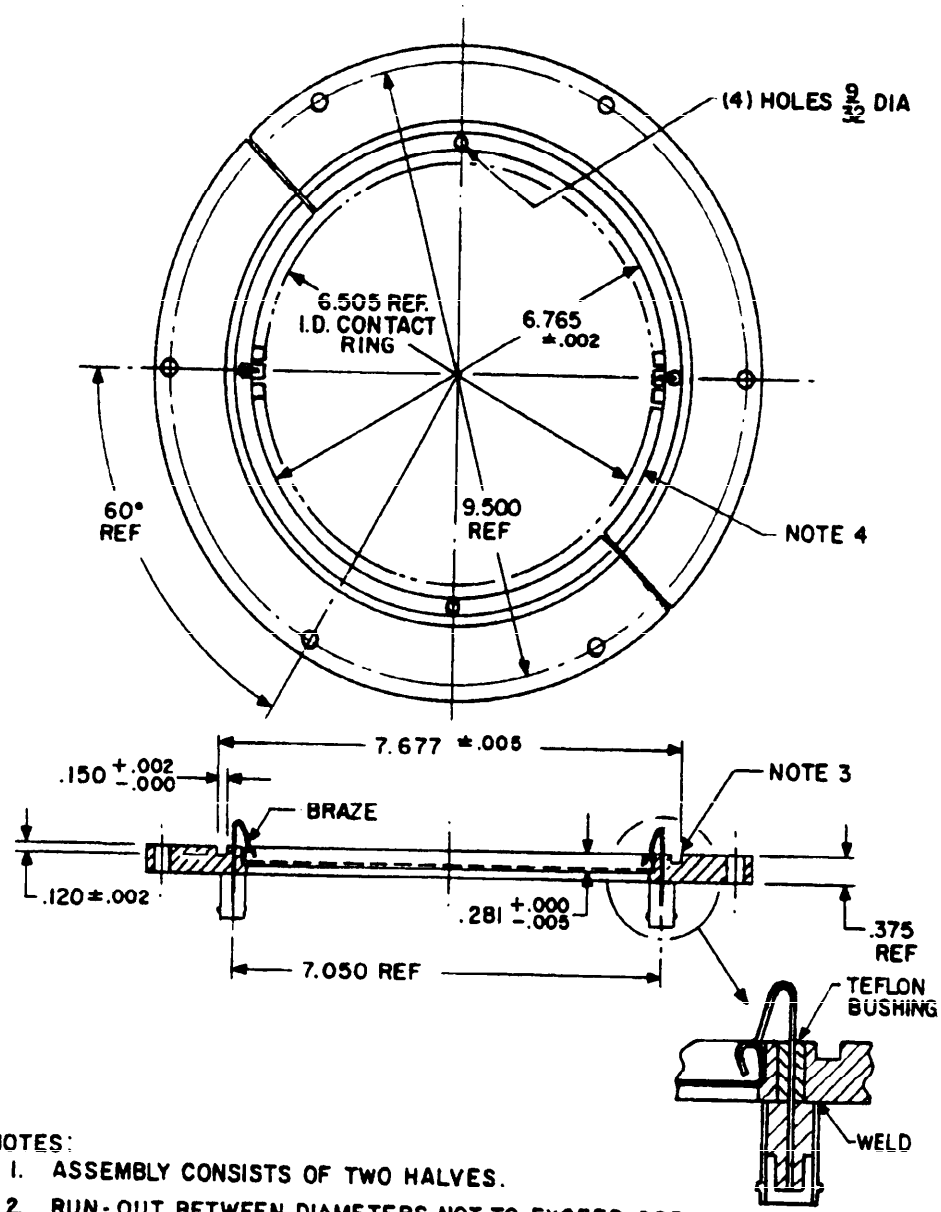
FIGURE 3. Lower Input Cavity (Inner)

NOTE:-

1. MOTE FOR BRAID CONTACT. THE BRAIDED RING, SUCH AS CAT. NO. 10-118, ELECTRONICS DIVISION, METAL TEXTILE CORP., ROSELLE N.J. (CODE 82805). MUST INSURE CONTINUOUS CONTACT.
2. DIMENSIONS ARE IN INCHES.

FIGURE 4. Upper Input Cavity

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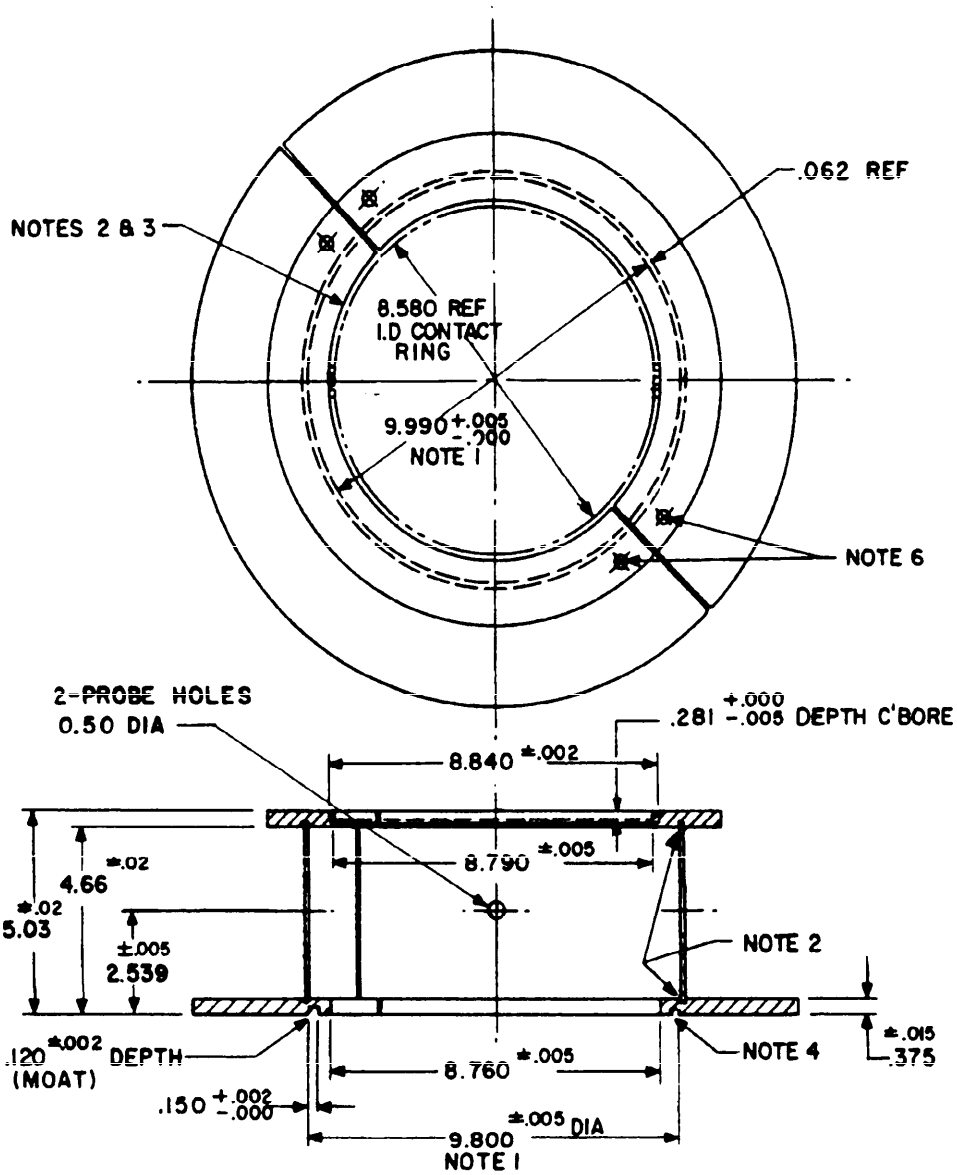


NOTES:

1. ASSEMBLY CONSISTS OF TWO HALVES.
2. RUN-OUT BETWEEN DIAMETERS NOT TO EXCEED .002
3. MOAT FOR BRADED CONTACT. THE BRAIDED RING, SUCH AS CAT. NO. 10-118, ELECTRONICS DIVISION, METAL TEXTILE CORP., ROSELLE, N.J. (CODE 82805)
4. SPRING CONTACT RING, SUCH AS CAT. NO. 97-135, INSTRUMENT SPECIALITIES CORP., LITTLE FALLS, N. J. (CODE 30817), MUST INSURE CONTINUOUS CONTACT.
5. DIMENSIONS ARE IN INCHES

FIGURE 5 Lower Input Cavity (Outer)

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NOTES:-

1. RUN-OUT BETWEEN DIAMETERS INDICATED NOT TO EXCEED .002
2. JUNCTURE MUST PROVIDE GOOD RF AND ELECTRICAL CONTACT AROUND ITS ENTIRE CIRCUMFERENCE, SUCH AS, BY BRAZE OR SOFT SOLDER CONNECTION. MAXIMUM BRAZE OR SOLDER FILLET NOT TO EXCEED .030 RADIUS.
3. SPRING CONTACT RING, SUCH AS CAT. NO. 97-135, INSTRUMENT SPECIALTIES CORP., LITTLE FALLS, N.J. (CODE 30817). MUST INSURE CONTINUOUS CONTACT.
4. MOAT FOR BRAIDED CONTACT. THE BRAIDED RING, SUCH AS CAT. NO. 10-118 ELECTRONICS DIV., METAL TEXTILE CORP., ROSELLE, N.J. (CODE 82805) MUST INSURE CONTINUOUS CONTACT.
5. ASSEMBLY CONSISTS OF TWO HALVES
6. TWO SMALL PLATES WITH MATCHING HOLES SHOULD BE USED TO HOLD HALVES TOGETHER
7. DIMENSIONS ARE IN INCHES.

FIGURE 6. Output Resonant Cavity

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